

DYO After Effects Video Agent

Complete Progress Handoff and Recovery Record

Status cutoff 16 September 2026

Current live project Mixkit Smartphone Promo Final

Prepared for Fahad and the next development session

This document consolidates the project documents and the material decisions from the working chat. It records what is complete, what failed, which repairs were deployed, the exact current state, and the remaining route to MVP acceptance.

Current conclusion: the control plane, Windows worker, AE/MCP health, safe working-copy recovery, and capture-only retry are proven. The First Preview must not be approved yet because the replacement screenshot and logo are not visible. A read-only working-copy card inspection is authorized and its result is still pending.

Security note

No passwords, database credentials, worker tokens, API keys, or registration secrets are included.

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1 Executive status

The project is well beyond initial setup. Production API and dashboard services, PostgreSQL persistence, the outbound Windows worker, worker identity reuse, AE and MCP health reporting, secure job dispatch, template inspection, plan approval, safe project copying, deterministic edit execution, checkpoints, and preview artifact upload all exist and have been exercised on the live system.

The current test reached Step 5 of 7, First Preview. All nine approved operations were checkpointed and the working copy was saved safely. A failed capture was recovered without repeating those edits. The recovery job succeeded and preserved both source and working-copy hashes. The remaining problem is visual: captures at 0.00 seconds and 2.00 seconds show a plain dark phone screen instead of the uploaded screenshot and logo. First Preview approval is therefore blocked pending inspection and correction of the image-replacement render path.

Area	Current status	Meaning
Control plane	GREEN	API, dashboard, database, jobs, approvals and worker registry are operational.
Windows worker	GREEN	Build 05dc0e3 installed; one process tree; stable online heartbeats.
After Effects and MCP	GREEN at last check	AE online, bridge live and MCP online during the latest preview run.
Approved edits	GREEN with evidence	All 9 operations recorded complete; saved working copy has a verified hash.
Capture-only recovery	GREEN	Exactly one recovery job ran without rebuilding, saving or reapplying edits.
First Preview visual	RED	At 0 s and 2 s the replacement screenshot and logo are not visible.
First Preview approval	BLOCKED	Do not approve or reject until the edited cards are inspected and a correct frame is produced.
Final Preview and export	NOT TESTED	Full preview, landscape render and native Reels remain locked behind First Preview.
MVP acceptance	PENDING	Must pass the complete workflow on three distinct plugin-free templates.

Immediate next action

Complete the authorized read-only inspection of App Screen 01 and App Screen 02 in the trusted working copy. The inspection must determine whether the imported images are hidden, covered, disabled, outside their timing window, assigned to the wrong source, affected by a track matte, or placed/scaled incorrectly. No mutation should occur until the root cause and a safe repair are reported.

2 Current live snapshot

Project: Mixkit Smartphone Promo Final

Dashboard stage: Step 5 of 7 First Preview

Session: 1257ac95-cebc-4980-8bb1-bbaaa2b2eac4

Session state: AWAITING_PREVIEW_APPROVAL

Plan: Revision 4 APPROVED

Approved operations: 9 operations indices 0 through 8

Working copy path: C:\DYO-Agent\execution-sessions\1257ac95-cebc-4980-8bb1-bbaaa2b2eac4\working-copy.aep

Working copy SHA 256: f38d5b258fbc627dcfe9c09c3e7c7b39cd6155d719fc4331d66d91ce08fad3f6

Immutable source SHA 256: 4172ee082c1ce15f9e180d17e128434d4b54a40b3596402e1a30e496415deafd

Worker build: 05dc0e30e3e37b9d23a47692b36492bb62c888aa

Worker process at validation: PID 13432 with one launcher supervisor worker process tree

Latest verified health: Worker ONLINE AE ONLINE MCP ONLINE bridge live

Active jobs after 2 s capture: None

Current operator rule: Do not approve reject regenerate or create complete preview until the card inspection result is reviewed

Evidence from the dashboard

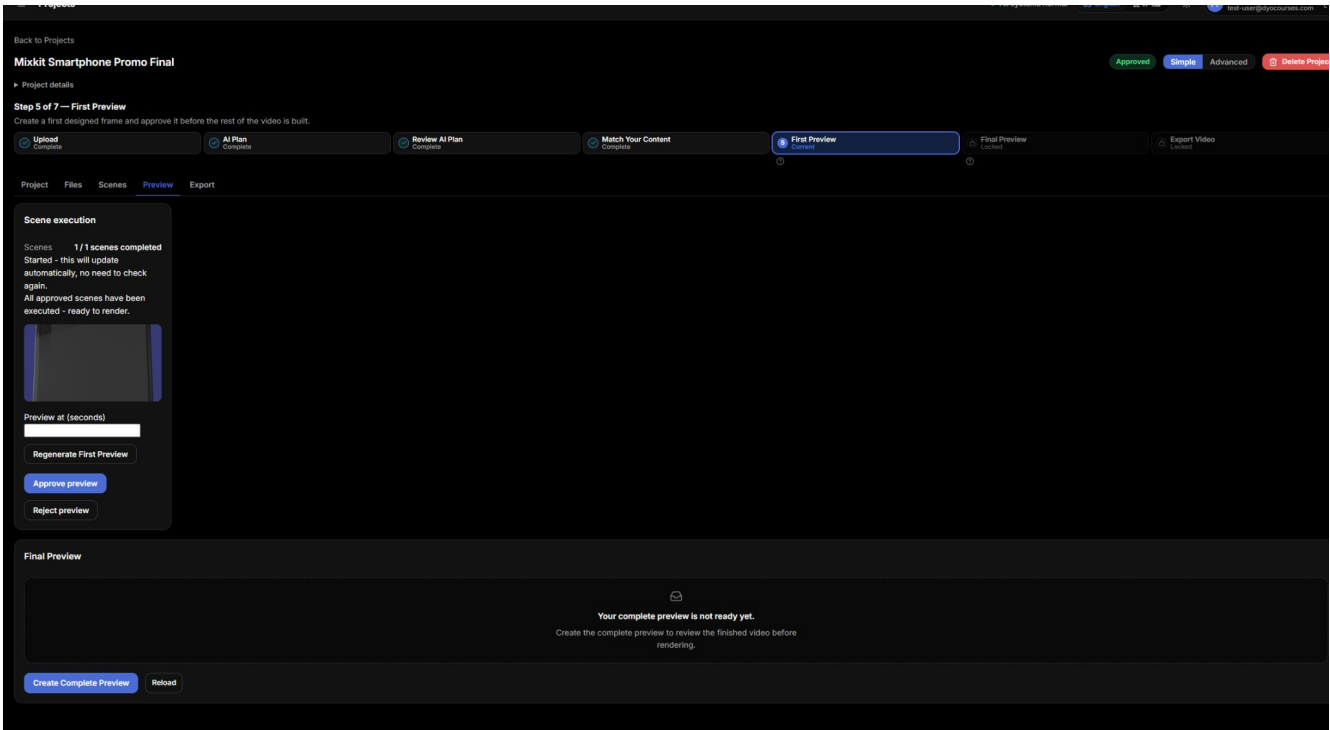


Figure 1 First Preview page after capture-only recovery. The UI reports 1 of 1 scenes completed, but the stored frame does not show the required edited content.

Current hold points

- Do not click Approve preview. The visual evidence is incorrect.
- Do not click Reject preview. That would mark the session FAILED without diagnosing the defect.
- Do not click Create Complete Preview. The first-frame gate has not passed.
- Do not regenerate again until the read-only card inspection reports the root cause or a specific diagnostic timestamp.
- Do not restart After Effects or the worker while inspection and evidence reconciliation are in progress.

3 Product and architecture

DYO is a deterministic After Effects automation system for adapting existing Envato or client templates into branded app-showcase videos. It is not a generic AI video generator. Human approval remains mandatory at the scene plan, first designed frame, branding, complete preview and final render gates.

Locked architecture

Component	Location	Responsibility
Web dashboard	Contabo	Guided project flow, mappings, approvals, previews, renders and operator diagnostics.

Component	Location	Responsibility
API	Contabo	Authentication boundary, worker registry, scheduling, job state, checkpoints and artifact metadata.
PostgreSQL	Contabo	Source of truth for workers, sessions, plans, jobs, approvals and artifacts.
Windows worker	Client PC	Outbound authenticated job claim, allowlisted operations, hashes, logs and checkpoints.
After Effects 2026	Client PC	Template inspection, deterministic edits, preview capture and final rendering.
ae-mcp bridge	Client PC	Controlled bridge between worker operations and After Effects JSX.
FFmpeg and FFprobe	Worker toolchain	Media inspection and exact source-frame extraction where required.
aerender	Client PC	Separate final rendering stage after approval.

Non negotiable safety rules

- Never overwrite the original AEP. Work from a copy and verify the source hash.
- Execute only versioned allowlisted operations. Never accept arbitrary shell PowerShell or JSX from a job.
- Keep edit operations and final rendering separate. Do not render inside an edit undo group.
- Use one AE job at a time on the current worker with maxConcurrency equal to 1.
- Checkpoint after meaningful operations and resume only from a verified safe checkpoint.
- Stop on AE modal MCP loss hash mismatch or uncertain saved state. Do not blindly retry destructive work.
- Actual AE previews from the exact output composition are required. Metadata-only success is not acceptance.
- Reels output must be a true 1080 by 1920 composition with elements repositioned, not a crop of landscape.

4 Work completed

Platform and deployment

- **COMPLETED** Contabo production services established for API, web dashboard and PostgreSQL.
- **COMPLETED** Public worker API protected through TLS and limited to authenticated worker POST operations.

- COMPLETED Worker registry, persisted worker identity, capabilities, max concurrency and heartbeat reporting implemented.
- COMPLETED Database-backed job claim, execution, checkpoints, reports and artifact metadata implemented.
- COMPLETED Simple and Advanced dashboard modes, project workflow stages, plan review and approval gates implemented.
- COMPLETED Final Preview proxy routes and backend preview artifact plumbing were implemented in the earlier redesign.

Windows worker and health

- COMPLETED One-time registration and persisted identity reuse proven.
- COMPLETED Automatic startup and supervisor recovery implemented.
- COMPLETED UTF-8 BOM environment-file bug fixed.
- COMPLETED AE path detection and AE ONLINE reporting fixed.
- COMPLETED ae-mcp instance discovery corrected to use the dot-prefixed data root and live instance evidence.
- COMPLETED Startup bridge recovery and bounded health probing added so normal restarts do not require daily manual connection.
- COMPLETED Build 05dc0e3 installed from a hash-verified release ZIP and confirmed as exactly one worker process tree.
- COMPLETED Nineteen consecutive ONLINE AE ONLINE MCP ONLINE heartbeats observed over five minutes before recovery execution.

Template and planning pipeline

- COMPLETED Read-only template inspection contract, stable IDs and manifest schema implemented.
- COMPLETED Dynamic plan and mapping flow implemented with human approval before AE mutation.
- COMPLETED Real effect and plugin scanning replaced the earlier empty pluginReferences stub.
- COMPLETED Version conversion path proven for an AE 23.2.1 project opened in AE 26.3 while preserving the original file.
- COMPLETED A converted template manifest was produced with 51 compositions, 50 nested compositions and top-level composition !Render at 1080 by 1920, 29.97 fps and about 45.045 seconds.
- COMPLETED Multiple earlier Envato candidates were correctly blocked because Element 3D was detected. A plugin-free acceptance set is still required.

Current project execution

- COMPLETED Upload, AI Plan, Review AI Plan and Match Your Content stages completed for the current project.
- COMPLETED Plan revision 4 approved.
- COMPLETED All nine approved operations completed and were checkpointed.
- COMPLETED Working copy saved to the execution session folder and hashed from disk.
- COMPLETED Immutable source re-hashed and confirmed unchanged.

- **COMPLETED** Capture-only resume logic deployed and proven on exactly one successful recovery job.
- **COMPLETED** First Preview artifacts at 0.00 seconds and 2.00 seconds were captured, uploaded and hash-verified.

5 Detailed chronological record

Stage	Verified record
Foundation	Architecture, security rules, server stack, database, worker registry, dashboard and phase plan were defined. After Effects remained the primary renderer after the Shotstack side proof could not reproduce AE phone and 3D behavior.
Worker pairing and startup	The real Windows worker registered, persisted its identity and began sending stable heartbeats. Scheduled startup and recovery scripts were added.
Environment repairs	BOM handling, worker exit propagation, AE_PATH persistence and ae-mcp data-root discovery were fixed. The target became Worker ONLINE, AE ONLINE and MCP ONLINE without manual daily setup.
Read-only inspection	Template inspector contracts and secure INSPECT_TEMPLATE dispatch were implemented. Plugin scanning later found Element 3D in several templates and prevented false plugin-free acceptance.
Template conversion	The dro template required AE 23.2.1 to 26.3 conversion. The original remained unchanged until a new Save As copy was deliberately created.
Dashboard redesign	Simple and Advanced flows, scene cards, preview plumbing and approval gates improved. Live testing still showed incorrect scene grouping and some manual preview behavior.
Current project plan	Mixkit Smartphone Promo Final reached plan revision 4 APPROVED with one selected scene and nine execution operations.
First execution	The worker applied operations 0 through 8 and saved the working copy. Final capture timed out while the bridge later appeared down. The UI misleadingly said it could not dispatch even though execution reached capture.
Recovery diagnosis	The saved working copy and source hash were verified. A plain retry was found unsafe because database key order made the old JSON.stringify comparison fail and

Stage	Verified record
	returned an empty resume checkpoint.
Recovery implementation	Commit 05dc0e3 added order-insensitive operation comparison, saved-working-copy hash validation, capture-only resume, longer bounded capture timeouts and a process-count correction.
Controlled rollout	API was deployed first. A read-only dry run initially failed because top-level await was compiled as CommonJS; the diagnostic script was fixed with async main and rerun successfully.
Checkpoint repair	A guarded idempotent SQL repair added the proven saved hash to job 60fcab36. UPDATE 1 and COMMIT were observed. The next dry run carried operations 0 through 8 and the exact saved hash.
Worker update	The release ZIP hash matched. The updater preserved registration and configuration, replaced worker files, verified dependencies and started exactly one process tree.
Capture-only proof	After five minutes of stable health, Continue execution was clicked once. Job 660a7690 succeeded with zero operations. The working copy and source hashes stayed unchanged.
Visual verification failure	The t=0 frame contained no edited content. Read-only timing inspection showed Smartphone_01 on screen from 0 to 4 seconds and Smartphone_05 from 16.04 to 21.44 seconds.
Second capture	The preview was regenerated once at 2.00 seconds. Job c43e5bf2 succeeded preview-only with zero operations and unchanged hashes, but the screenshot and logo were still absent.
Present checkpoint	Read-only inspection of App Screen 01 and App Screen 02 in the trusted working copy was authorized. Its result and any proposed fix have not yet been received.

6 First Preview recovery incident

Initial failure

The original execution completed all nine edit operations and reached final preview capture. The ae_capture_frame operation returned MCP error -32001 after its short timeout. No preview file

reached the worker, and the bridge later reported zero live instances. The saved working copy nevertheless existed on disk and had been hashed after operation 8.

Evidence	Value
Original failed job	60fcab36-f116-4277-96dd-72d9d30c79af
Completed operations	0 1 2 3 4 5 6 7 8
Working-copy hash	f38d5b258fbc627dcfe9c09c3e7c7b39cd6155d719fc4331d66d91ce08fad3f6
Source hash	4172ee082c1ce15f9e180d17e128434d4b54a40b3596402e1a30e496415deafd
Capture artifact	None from the failed attempt
Unsafe old behavior	A retry could rebuild the copy and reapply all 9 operations because the resume comparison failed on JSON key order.

Implemented repair

- Resume only when every operation completed and a verified saved working-project hash exists.
- Require the existing working copy to match the saved hash. Refuse missing or different files.
- For a valid completed checkpoint, reopen the saved copy and perform capture only. Apply and save nothing.
- Use 60 seconds for connection and 180 seconds for capture. Do not automatically queue a second render.
- Compare operation content without depending on database JSON key order.
- Correct the installer process-count check so zero processes cannot be treated as one.

Controlled database repair and validation

The checkpoint repair was constrained to the exact failed job, nine completed operations, the known capture failure and the proven working-copy hash. It was idempotent and updated one row. A dry run then proved that Continue would carry the full checkpoint rather than null.

```
BEGIN
UPDATE 1
COMMIT
```

REAL resolver result:

completedOperationIndices = [0,1,2,3,4,5,6,7,8]

savedWorkingProjectSha256 = f38d5b258fbc627dcfe9c09c3e7c7b39cd6155d719fc4331d66d91ce08fad3f6

Successful capture-only execution

Check	Result
Job	660a7690 only job on the session after Continue
Time	15:13:31 to 15:13:41 UTC on 15 September 2026
Worker	PID 13432 build 05dc0e3 one process tree
Operations applied	Zero
Rebuild or save	None
Working copy	Hash unchanged
Source	Hash unchanged
Preview	Valid 640 by 360 PNG attached to the session
Session after job	AWAITING_PREVIEW_APPROVAL

7 Current preview rendering defect

Observed result

The capture mechanism now works, but the rendered visual does not prove the requested edits. The uploaded screenshot is a dark navy Vehicles app screen with bright white text and red and purple icons. The uploaded logo is a red SIM-card icon on light gray. Neither appears in the t=0 or t=2 frame. Both frames show plain dark placeholder surfaces.

Measured timing

Content	Composition	Main Comp window	Diagnostic time
Screenshot and logo	Smartphone_01	0.00 to 4.00 s	2.00 s captured and failed visual check
Hebrew text and second screenshot	Smartphone_05	16.04 to 21.44 s	19.00 s proposed but not yet captured
Hebrew text layer	Smartphone_05	Starts 0.40 s into phone	Cannot appear before 16.44 s

Smartphone_01 and Smartphone_05 are not on screen together, so one frame cannot prove every edit. Two accepted diagnostic frames may be required. However, the 19-second capture should remain on hold until the missing image rendering in Smartphone_01 is understood.

Latest 2 second capture evidence

Check	Result
Job	c43e5bf2 only execution job since the timing inspection
Execution	Preview-only zero operations succeeded
Time	15:36:18 to 15:36:24 UTC on 15 September 2026
Working copy	Hash unchanged
Source	Hash unchanged
Preview artifact	Valid 640 by 360 PNG stored SHA begins 87b9a44d
Visual	Central phone and background cards remain plain dark gray; no screenshot details and no red logo are visible

Authorized inspection

The next read-only inspection is authorized for App Screen 01 and App Screen 02 in the trusted working copy. It should report layer order, enabled state, opacity, in and out times, assigned source, track matte, scaling and position. The inspection must verify the working-copy hash before opening and must save nothing. The developer must report the proven root cause and a safe fix before any mutation.

8 Known bugs and fixes

Issue	Observed cause or symptom	Resolution or status
Windows env BOM	DYO_API_URL appeared undefined because PowerShell 5.1 wrote a UTF-8 BOM.	Write UTF-8 without BOM and validate with Node.
Worker restart masking	A formatter pipeline hid the real worker exit code.	Raw background wrapper now propagates the worker exit code.
AE health unknown	AE_PATH was not persisted.	Persist discovered AfterFX.exe path and validate it.
MCP path wrong	Worker looked under ae-mcp without the dot-prefixed runtime data root.	Discover valid live instances under %USERPROFILE%\ae-mcp\instances*.
MCP probe timeout	Short probe produced UNKNOWN while bridge initialization was still progressing.	Use bounded startup recovery and reconcile real bridge health.
Plugin scan false negative	pluginReferences was effectively	Real effect scan across all

Issue	Observed cause or symptom	Resolution or status
	empty.	compositions added.
Template conversion	AE 23.2.1 template required conversion in AE 26.3.	Open in memory and Save As a separate new file; never overwrite source.
Capture timeout	ae_capture_frame timed out at the end of a successful nine-operation execution.	Capture-only resume with 60-second connect and 180-second capture timeout.
Resume key order	JSON.stringify comparison failed when database reordered object keys.	Order-insensitive comparison plus verified saved hash.
Installer count	Zero process results could be misread as exactly one.	Correct process counting and verify one process tree.
Dry-run CommonJS error	Top-level await caused eight TransformErrors.	Wrap the script in async main with guarded cleanup.
Misleading dispatch UI	UI said Could not dispatch even though work reached capture.	Recorded as acceptance issue; UI reconciliation still needed.
Preview time default	t=0 can show no edited content.	Use composition timing evidence and content-aware representative timestamps.
Image replacement visual	Worker reports success but screenshot and logo do not render in the preview.	Root cause pending authorized read-only card inspection.

9 Work remaining

Immediate current-project work

- **PENDING** Complete read-only working-copy card inspection and record evidence.
- **PENDING** Identify the exact cause of the missing screenshot and logo.
- **PENDING** Implement a bounded safe repair only after root cause review and authorization.
- **PENDING** Regenerate a Smartphone_01 frame and verify screenshot plus logo visually.
- **PENDING** Capture a Smartphone_05 frame around 19.00 seconds and verify Hebrew text plus second screenshot.
- **PENDING** Approve First Preview only after required edits are visually proven.
- **PENDING** Generate, play and approve the complete Final Preview.
- **PENDING** Run final landscape render, download it and play it end to end.
- **PENDING** Run true native 1080 by 1920 Reels render and verify it is repositioned rather than cropped.

Product and acceptance work

- PENDING Finish evidence-based real AE nesting and source-composition detection where still incomplete.
- PENDING Regenerate affected manifests and ensure Simple Mode shows real scenes rather than technical nested compositions.
- PENDING Make missing and stale real-AE previews automatic and self-updating without manual browser refresh.
- PENDING Fix and live-test INSPECT_RENDER_CAPABILITIES if the JSON is undefined issue still exists in the deployed build.
- PENDING Run an intentional interrupted Worker or AE recovery test without project corruption or destructive retry.
- PENDING Repeat the complete workflow on two additional distinct plugin-free templates.
- PENDING Archive screenshots, artifact hashes, outputs and PASS FAIL evidence for all three templates.
- PENDING Rotate exposed or temporary credentials and remove obsolete packages only after live debugging ends.

Current stage estimate

The current project is at First Preview, not Final Preview. Infrastructure and safe recovery are largely complete, but video acceptance is not. The remaining scope includes the image-replacement fix, two visual checks, First Preview approval, complete preview, both final formats, recovery acceptance and two more plugin-free template runs. The project must therefore remain marked MVP acceptance pending.

10 Safe resume instructions

First message for the next Claude Code session

Continue the current Mixkit Smartphone Promo Final session from the verified First Preview checkpoint.

Read this handoff before taking action. Do not approve, reject, regenerate, render, restart AE, restart the worker, or mutate either AEP until live state is reconciled.

Verify:

- session 1257ac95-cebc-4980-8bb1-bbaaa2b2eac4
- state AWAITING_PREVIEW_APPROVAL
- plan revision 4 APPROVED
- no active jobs
- worker build 05dc0e3 and exactly one process tree
- Worker ONLINE, AE ONLINE, MCP ONLINE and bridge live
- working-copy SHA-256 f38d5b258fbc627dcfe9c09c3e7c7b39cd6155d719fc4331d66d91ce08fad3f6
- source SHA-256 4172ee082c1ce15f9e180d17e128434d4b54a40b3596402e1a30e496415deafd

The t=0 and t=2 previews are valid artifacts but show neither the uploaded screenshot nor the logo. A read-only inspection of App Screen 01 and App Screen 02 was authorized. Complete that inspection only, report layer order/enabled/opacity/timing/source/track matte/scale/position, identify the proven root cause, and propose a bounded safe fix. Save nothing and wait for approval before mutation.

Operator decision sequence

1. Reconcile live database job session and artifact state.
2. Reconfirm both hashes and health.
3. Complete the read-only card inspection.
4. Stop and report the root cause with evidence.
5. Obtain explicit approval for any mutation.
6. Apply only the bounded fix to the trusted working copy with a new checkpoint and hash.
7. Capture and inspect Smartphone_01, then Smartphone_05.
8. Approve First Preview only when visual evidence passes.
9. Proceed to complete preview and exports under existing approval gates.

11 Acceptance checklist

Acceptance gate	Status	Evidence or requirement
Production health green	PARTIAL PASS	Passed during current recovery; recheck after every deploy or restart.
Original AEP unchanged	PASS current run	Source hash remained 4172ee082...deafd.
Working copy protected	PASS current run	Hash-based capture-only resume refused rebuild or mismatch.
Approved operations executed	PASS current run	Nine operations checkpointed.
First Preview visual correct	FAIL	Screenshot and logo absent at 0 s and 2 s.
Hebrew text visual correct	NOT TESTED	19 s diagnostic capture not yet run.
First Preview approved	NO	Intentionally held.
Complete Final Preview	NOT TESTED	Locked behind First Preview.
Landscape MP4	NOT TESTED	Must render download and play.
Native Reels MP4	NOT TESTED	Must prove real 1080 by 1920 repositioning.
Interrupted-job recovery	PARTIAL PASS	Capture-only recovery proven; final intentional acceptance scenario still required.
Three plugin-free templates	FAIL	Current full acceptance set not complete.

Release rule

The MVP may be called complete only when every mandatory gate passes on three distinct plugin-free templates. A green subsystem does not make the whole product complete. The correct present label is MVP acceptance pending.

12 Commands and evidence

Read-only dry run used for resume validation

```
cd /opt/AF-agent/apps/api && DATABASE_URL="$(grep ^DATABASE_URL= /opt/AF-agent/.env | cut -d= -f2- | tr -d '"")"
npx tsx /tmp/claude-1002/-opt-AF-agent/0f4cbfb5-2f5f-4cc6-8c0d-2169774d3cf2/scratchpad/dry-run-resume.ts
```

The dry run initially returned null because the saved hash was absent. After the guarded repair, it returned operations 0 through 8 and the exact saved working-copy hash. The npm upgrade notice was unrelated and harmless.

Guarded checkpoint repair used once

```
psql "$(grep ^DATABASE_URL= /opt/AF-agent/.env | cut -d= -f2- | tr -d '"")" -f /tmp/claude-1002/-opt-AF-agent/0f4cbfb5-2f5f-4cc6-8c0d-2169774d3cf2/scratchpad/repair-60fcab36-saved-hash.sql
```

Observed result: BEGIN, UPDATE 1, COMMIT. This was a one-time constrained production repair, not a normal operating step.

Worker release evidence

Release file: DYO-QA-Worker-CaptureOnlyResume.zip

Release SHA 256: 13E77BEB5465D41B7F431D9E950AEAA368ECA4664F5EDC78D03793D92DCAD91

Installed build: 05dc0e30e3e37b9d23a47692b36492bb62c888aa

The installer preserved the existing worker registration and configuration, verified runtime dependencies and started exactly one process tree.

13 Source document index

Source	Contribution
DYO AE Video Agent Client Guide	Client-facing project flow, approval gates, preview and export behavior.
DYO Claude Code Development Brief	Locked stack, phase plan, safety constraints and acceptance requirements.
Complete Guide	Dashboard UX review, live acceptance checklist and known open product issues.
Master Handoff Recovery 25 August 2026	Infrastructure history, worker deployment, recovery rules and rebuild guidance.

Source	Contribution
README START HERE	Original continuation instructions.
CLAUDE	Mission, architecture, brand rules, workflow and safety rules.
PHASES	Implementation phases 0 through 10.
MASTER PLAN	Final architecture, lifecycle, operations and acceptance design.
ARCHITECTURE	Component responsibilities, scheduler and security boundaries.
CLIENT WORKER PREFLIGHT	Read-only Windows readiness checks.
SCHEMAS	Manifest, execution plan and brand-rule data contracts.
RUNBOOK	Operational sequence and safe recovery behavior.
ACCEPTANCE	Minimum evidence required before MVP completion.
DEFINITION OF DONE	Release completion standard.
Current working chat	Latest capture-only recovery, preview timing analysis, visual defect and authorized inspection state.

14 Consolidated chat record

This section preserves the operational substance of the recent chat in chronological order. Repeated confirmations and profanity were omitted; every material instruction, result, decision, identifier and safety stop was retained.

Step	Chat event	Outcome
1	Operator asked how much remained.	First Preview was assessed at about 90 to 95 percent; capture was the known blocker, while Final Preview and Export remained untested.
2	Claude verified the failed execution.	All 9 operations completed; working copy saved and hashed; source unchanged; bridge down; no capture file.
3	Unsafe retry discovered.	Every retry had an empty checkpoint because stored operation keys reordered and the old JSON.stringify comparison

Step	Chat event	Outcome
		failed.
4	Fix prepared.	Commit 05dc0e3 added capture-only resume, hash validation, longer capture timeouts and process-count correction.
5	API deployed first.	Production API restarted successfully with the new resume check.
6	First dry run failed.	The diagnostic script used top-level await under CommonJS and produced 8 TransformErrors. No database work occurred.
7	Dry run repaired.	Script moved to async main with error handling and guaranteed database cleanup.
8	Pre-repair dry run passed.	Old check false, semantic operation match true, real resolver null, as expected without saved hash.
9	Checkpoint repaired.	Guarded SQL produced UPDATE 1 and COMMIT with the exact saved hash.
10	Post-repair dry run passed.	Resolver carried operations 0 through 8 and the exact saved hash.
11	Bridge health confirmed.	Bridge already live, one instance, working copy open; no manual reconnect needed.
12	Worker update installed.	ZIP hash True; registration preserved; dependencies verified; exactly one process tree started.
13	Five-minute stability gate passed.	19 consecutive ONLINE AE ONLINE MCP ONLINE heartbeats; bridge passing; no active jobs.
14	Continue clicked once.	Monitor watched for exactly one capture-only execution.
15	Capture-only job succeeded.	Job 660a7690 applied zero operations; both hashes stayed unchanged; valid preview uploaded.

Step	Chat event	Outcome
16	t=0 frame rejected as evidence.	It showed a blank dark phone surface and none of the screenshot, logo or Hebrew text.
17	Timing inspection authorized.	Read-only Main Comp timing showed Smartphone_01 from 0 to 4 s and Smartphone_05 from 16.04 to 21.44 s.
18	Regenerate at 2 s clicked once.	Job c43e5bf2 succeeded preview-only with zero operations and unchanged hashes.
19	t=2 frame also failed visual check.	Screenshot and logo were still absent, indicating a rendering or replacement problem rather than only a timestamp problem.
20	Working-copy card inspection authorized.	Read-only inspection of App Screen 01 and App Screen 02 may proceed; no repair is authorized yet.

End state

No approval or rejection has been given for First Preview. No complete preview or final render has started. The trusted working copy and source hashes remain the central safety anchors. Resume from the authorized read-only card inspection and do not repeat completed edits.